WORKED SOLUTION

STEP II 1992, Question 10

Pure • Loci in the Argand Diagram

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q10

The key idea

Write $\dfrac{z-1}{z} = \dfrac{1-z}{0-z}$: its argument is the angle, turned anticlockwise, from the direction $z \to 0$ to the direction $z \to 1$. So fixing the argument fixes the angle that the segment from 0 to 1 subtends at z, and circle theorems (angles in the same segment, opposite angles of a cyclic quadrilateral) take over. The final part is the same idea with the chord z_4z_2 in place of 0 to 1.

Solution

(The statement of the question has a misprint: the condition is meant to be $0 < \alpha \leq \frac{1}{2}\pi$.)

Two facts do all the work. First, since $\arg\dfrac{z-1}{z} = \arg(1-z) - \arg(0-z)$, the argument measures the angle at z from the line $z0$ round to the line $z1$, anticlockwise. Second, it tells us which side of the real axis z is on:

$$\operatorname{Im}\frac{z-1}{z} = \operatorname{Im}\left(1 - \frac{1}{z}\right) = \operatorname{Im}\left(-\frac{\bar{z}}{|z|^2}\right) = \frac{\operatorname{Im} z}{|z|^2}.$$

So $\arg\frac{z-1}{z} \in (0, \pi)$ exactly when $\operatorname{Im} z > 0$, and lies in $(-\pi, 0)$ exactly when $\operatorname{Im} z < 0$. Points on the real axis give a real quotient (argument 0 or π), and $z = 0, 1$ give no argument at all.

Part (i)

Here $\operatorname{Im} z > 0$, and the segment from 0 to 1 subtends the angle α at z. The points above the real axis at which a fixed chord subtends a fixed angle form an arc of a circle through the chord's ends (angles in the same segment are equal). Conversely, every point of that arc sees the chord at angle α, turned anticlockwise because the point lies above the chord, so the whole arc belongs to the locus.

To pin the circle down: the angle at the centre is 2α, so a chord of length 1 gives radius

$\dfrac{1}{2\sin\alpha}$, and the centre lies on $\operatorname{Re} z = \frac{1}{2}$ at height $\frac{1}{2}\cot\alpha \geq 0$. The locus is the arc of

$$\left|z - \tfrac{1}{2} - \tfrac{\mathrm{i}}{2}\cot\alpha\right| = \frac{1}{2\sin\alpha}$$

lying above the real axis, with the end points 0 and 1 excluded. Since $\alpha \leq \frac{1}{2}\pi$ this is the major arc. When $\alpha = \frac{1}{2}\pi$ it is the upper semicircle on the diameter from 0 to 1 (the angle in a semicircle).

Part (ii)

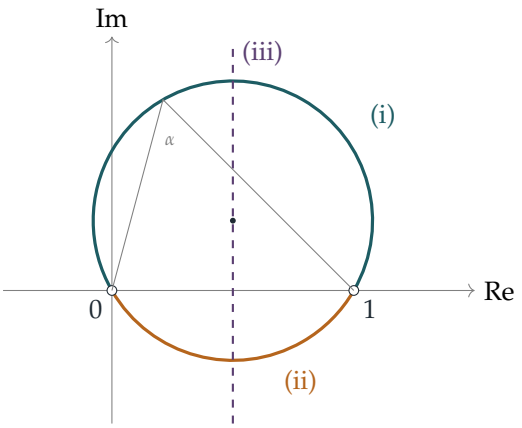
Now the argument $\alpha - \pi$ lies in $(-\pi, -\frac{1}{2}\pi]$, so $\operatorname{Im} z < 0$, and the chord subtends an angle of size $\pi - \alpha$ at z, turned clockwise.

Ask yourself Which circle has the chord subtending $\pi - \alpha$ below the axis?

The same circle as in (i). If P is on the arc from (i) and Q is on the other arc of that circle, then $0PQ1$ (taken in a suitable order) is a cyclic quadrilateral with P and Q opposite, so the angles at P and Q add up to π. Hence the minor arc below the axis is exactly where the angle is $\pi - \alpha$. The locus is the arc of the circle from (i) below the real axis, again without 0 and 1. Parts (i) and (ii) together make up the whole circle apart from 0 and 1. (If $\alpha = \frac{1}{2}\pi$, (ii) is the lower semicircle.)

Part (iii)

$\left|\frac{z-1}{z}\right| = 1$ means $|z-1| = |z|$ with $z \neq 0$: z is equidistant from 0 and 1. The locus is the perpendicular bisector, the line $\operatorname{Re} z = \frac{1}{2}$.



The sketch shows $\alpha = \frac{1}{3}\pi$: the teal major arc is (i), the amber minor arc is (ii), the dashed plum line is (iii), and the open circles mark the excluded points 0 and 1.

The final part

Let $f(z) = \dfrac{z - z_2}{z - z_4}$. Since $(z_1 - z_4)(z_3 - z_2) = (z_4 - z_1)(z_2 - z_3)$,

$$\frac{f(z_1)}{f(z_3)} = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_1 - z_4)(z_3 - z_2)} = \frac{(z_1 - z_2)(z_3 - z_4)}{(z_4 - z_1)(z_2 - z_3)} = w,$$

so $\arg w = \arg f(z_1) - \arg f(z_3) \pmod{2\pi}$.

The function f is the function of parts (i) and (ii) in new coordinates. Put $z = z_4 + (z_2 - z_4)u$, which is a rotation and enlargement followed by a translation, taking $u = 0, 1$ to z_4, z_2. Then

$$f(z) = \frac{(z_2 - z_4)(u - 1)}{(z_2 - z_4)u} = \frac{u - 1}{u}.$$

The circle through the four points corresponds to a circle in the u-plane through 0 and 1.

? **Ask yourself** Why are z_1 and z_3 on different arcs?

The points come in the order z_1, z_2, z_3, z_4 round the circle, so the chord z_2z_4 cuts the circle into two arcs, one containing z_1 and the other containing z_3. In the u-plane, one of them lies above the real axis and the other below it.

From (i) and (ii), the argument of $\frac{u-1}{u}$ takes a constant value $\beta \in (0, \pi)$ on the upper arc and the value $\beta - \pi$ on the lower arc. Here we are not assuming $\beta \leq \frac{1}{2}\pi$: the circle-theorem argument above never used that restriction. Therefore

$$\arg w = \arg f(z_1) - \arg f(z_3) = \pm\big(\beta - (\beta - \pi)\big) = \pm\pi \pmod{2\pi},$$

so w is real, and in fact negative. ■

Checks

For the square $z_1 = 1$, $z_2 = \mathrm{i}$, $z_3 = -1$, $z_4 = -\mathrm{i}$ we get $w = \dfrac{(1-\mathrm{i})(-1+\mathrm{i})}{(-1-\mathrm{i})(1+\mathrm{i})} = \dfrac{-(1-\mathrm{i})^2}{-(1+\mathrm{i})^2} = \dfrac{-2\mathrm{i}}{2\mathrm{i}} = -1$, which is real and negative. For (i), take $\alpha = \frac{1}{2}\pi$ and $z = \frac{1}{2} + \frac{1}{2}\mathrm{i}$. Then $\frac{z-1}{z} = \frac{-1+\mathrm{i}}{1+\mathrm{i}} = \mathrm{i}$, whose argument is $\frac{1}{2}\pi$. We also checked numerically that the arcs carry the stated arguments for several values of α, and that w was real and negative for 10^4 random sets of four points in order on random circles.

Common slips

- Giving the whole circle for (i). Only one arc has the right argument; the other arc gives $\alpha - \pi$, which is exactly part (ii).
- Forgetting to exclude 0 and 1, where the argument is undefined.

- Getting the sign of $\arg w$ wrong because the minus signs in $(z_4 - z_1)(z_2 - z_3)$ are mishandled. Pairing them as above avoids this.

Beyond STEP

The quantity w is (up to sign and ordering) the *cross-ratio* of four points. It is unchanged by every Möbius transformation $z \mapsto \frac{az+b}{cz+d}$, and four points lie on a circle or line exactly when their cross-ratio is real. Since Möbius maps send circles and lines to circles and lines, this gives a neat way to prove concyclicity. Needham's *Visual Complex Analysis* develops this geometric view beautifully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.